

Microbial and sensory changes throughout the ripening of Prato cheese made from milk with different levels of somatic cells.

The objective of this research was to evaluate the effects of 2 levels of raw milk somatic cell count (SCC) on the composition of Prato cheese and on the microbiological and sensory changes of Prato cheese throughout ripening. Two groups of dairy cows were selected to obtain low-SCC (<200,000 cells/mL) and high-SCC (>700,000 cells/mL) milks, which were used to manufacture 2 vats of cheese. The pasteurized milk was evaluated according to the pH, total solids, fat, total protein, lactose, standard plate count, coliforms at 45 degrees C, and *Salmonella* spp. The cheese composition was evaluated 2 d after manufacture. Lactic acid bacteria, psychrotrophic bacteria, and yeast and mold counts were carried out after 3, 9, 16, 32, and 51 d of storage. *Salmonella* spp., *Listeria monocytogenes*, and coagulase-positive *Staphylococcus* counts were carried out after 3, 32, and 51 d of storage. A 2 x 5 factorial design with 4 replications was performed. Sensory evaluation of the cheeses from low- and high-SCC milks was carried out for overall acceptance by using a 9-point hedonic scale after 8, 22, 35, 50, and 63 d of storage. The somatic cell levels used did not affect the total protein and salt:moisture contents of the cheeses. The pH and moisture content were higher and the clotting time was longer for cheeses from high-SCC milk. Both cheeses presented the absence of *Salmonella* spp. and *L. monocytogenes*, and the coagulase-positive *Staphylococcus* count was below 1×10^2 cfu/g throughout the storage time. The lactic acid bacteria count decreased significantly during the storage time for the cheeses from both low- and high-SCC milks, but at a faster rate for the cheese from high-SCC milk. Cheeses from high-SCC milk presented lower psychrotrophic bacteria counts and higher yeast and mold counts than cheeses from low-SCC milk. Cheeses from low-SCC milk showed better overall acceptance by the consumers. The lower overall acceptance of the cheeses from high-SCC milk may be associated with texture and flavor defects, probably caused by the higher proteolysis of these cheeses.